

Cross-Attention against Late, FiLM, and Gated Fusion for Caption-Guided Land-Cover Understanding

Süleyman Ceran

Middle East Technical University

Graduate School of Informatics

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I. LITERATURE SURVEY

Vision-language models (VLMs) jointly represent visual and textual information through contrastive learning [1] and have gained traction in remote sensing (RS), where Wen et al. [2] identify “cross-modal fusion” as a major open challenge. RemoteCLIP [3], the first CLIP for RS, outperforms vanilla CLIP by up to 6.39% on zero-shot classification via domain-adapted pretraining. Recent multimodal LLMs such as EarthGPT [4] and RingMoGPT [5] unify captioning, VQA, grounding, and change detection within a single instruction-tuned framework that projects visual tokens into LLM token spaces.

A central design choice is how to fuse the two modalities. Cross-modal attention has beaten unimodal methods on RS VQA [6] and visual grounding [7], and SegCLIP [8] fuses text with image features for RS segmentation; FiLM [9] offers an attention-free alternative via per-feature affine modulation. Yet few studies directly compare cross-attention against late or feature-wise fusion on RS scene-level tasks, and the effect of caption-generation strategy on fusion performance remains under-explored despite evidence that caption source and quality strongly affect multi-modal models [10].

II. PROJECT PROPOSAL AND PROOF OF CONCEPT

RQ (Phase-1): How effective is cross-attention compared to late fusion for RS scene classification with LLM-generated captions? **Dataset:** ARAS400K [11]—10,000 satellite images with 7-class segmentation masks, composition percentages, and 5 captions per image (vision-only, text-only, hybrid); Phase-1 used an 8-class dominant-class label (threshold 50%). **Method:** a frozen vanilla CLIP-ViT-B/32 backbone with two trainable heads—late fusion of pooled CLS embeddings and cross-attention with text-as-queries over image patches—across 5 captions $\times$ 2 fusions plus image-only (11 conditions). **PoC:** 81% val accuracy on a 500-sample, 5-epoch run (vs. 12.5% chance) confirmed pipeline feasibility.

III. METHOD REFINEMENT

The Phase-1 review [12] raised three concerns—trivial task, unjustified backbone, insufficient sample size—addressed along three corresponding axes. A pre-experiment confirmed the first: a text-only classifier on the original dominant-class task reaches mAP $\approx$ 0.83 alone, since captions encode the same composition statistics that define the labels. We re-cast the target as 7-bit multi-label class presence with threshold $\tau=10\%$, exposing minor classes (Shrub, Built-up, Water, Barren; each $<11\%$ of images) where captions are unreliable and image-text complementarity is genuine;

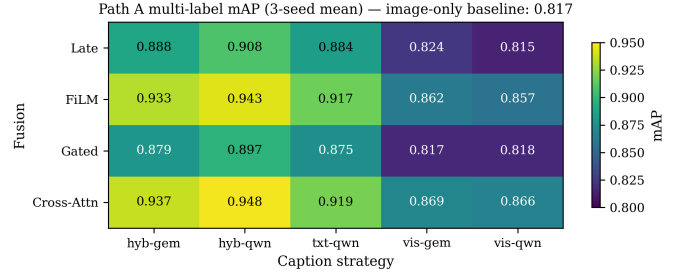


Fig. 1. Path A multi-label mAP (3-seed mean): fusion (rows) vs. caption (cols). Image-only baseline: 0.817 ± 0.001 .

we also pilot a patch-level segmentation extension (Path B, Sec. VI). The frozen backbone is upgraded to RemoteCLIP-ViT-B/32 [3], with forward hooks on `ln_post/ln_final` recovering all 50 patch tokens and 77 text tokens. The fusion comparison expands from 2 to 4: *Late* concatenation $[i||t]$; *FiLM* [9] ($i'=(1+\gamma) \odot i + \beta$, (γ, β) from pooled text); *Gated* ($g=\sigma(W[i||t])$, $g \odot i + (1-g) \odot t$); and *Cross-Attention* (CA) with multi-head attention—text tokens query image patches for classification, the direction *inverted* for segmentation. All share a 256-unit ReLU-Dropout-Linear head; only the fusion module and head train.

IV. EXPERIMENTAL SETUP

We use the full 10K ARAS400K corpus, an 80/20 stratified train/val split (stratifying on the dominant class, fixed seed), and 3 random seeds per condition. All conditions train for 30 epochs with AdamW ($\text{lr}=10^{-3}$, $\text{weight-decay}=10^{-4}$, batch 256/128) using BCE-with-logits (Path A) or per-patch cross-entropy (Path B). Best-epoch metrics are reported as mean $\pm$ std across the three seeds: mAP, macro F1, and per-class F1 for Path A; mIoU and per-class IoU for Path B. With features cached after a single forward pass through the frozen backbone, the entire 117-run matrix (33 Path A + 33 backbone + 30 fusion-variant + 21 seg) completes in <15 min on one A100. All runs are public on Weights & Biases and the codebase on GitHub [13], addressing the Phase-1 reproducibility feedback.

V. MULTI-LABEL CLASSIFICATION: RESULTS AND ABLATIONS

Headline. Cross-attention with `hybrid_qwen3-vl-8b` attains mAP 0.948 ± 0.003 and macro F1 0.867 ± 0.014 , against an image-only baseline of 0.817 ± 0.001 —an absolute +0.131 mAP gain from caption-guided fusion. Fig. 1 shows the full 4×5 matrix; Table I the per-family best.

CA dominates Late on every caption (Δ mAP = +0.036 to +0.051, std ≤ 0.005). The gap is largest with the noisy

TABLE I
BEST PATH A RESULT PER FUSION FAMILY (3-SEED MEAN $\pm$ STD).

Fusion	Best mAP	Macro F1	Mean mAP
Image-only	0.817 $\pm$ 0.001	0.702 $\pm$ 0.006	—
Late	0.908 $\pm$ 0.003	0.836 $\pm$ 0.009	0.864
Gated	0.897 $\pm$ 0.001	0.811 $\pm$ 0.007	0.857
FiLM	0.943 $\pm$ 0.001	0.876 $\pm$ 0.008	0.903
Cross-Attn	0.948 $\pm$ 0.003	0.867 $\pm$ 0.014	0.908

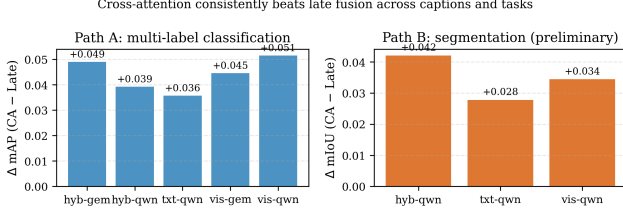


Fig. 2. CA-vs-Late advantage across captions, both tasks. Same pattern in classification (left, Δ mAP) and preliminary segmentation (right, Δ mIoU); largest gap on the noisy vision_qwen caption.

vision_qwen3-vl-8b: Late gains nothing over image-only ($0.815 \approx 0.817$), but CA recovers $+0.049$. Vision-only LLM captions frequently *hallucinate* (e.g. a 92%-grass scene labeled “barren”); CA’s attention filters unreliable tokens, while the alternatives cannot.

Family ranking (mean mAP across captions): CA $0.908 >$ FiLM $0.903 >$ Late $0.864 >$ Gated 0.857 . FiLM matches CA within 0.005 mAP at lower cost; Gated *underperforms* Late as the sigmoid gate collapses one modality (a negative result).

Backbone ablation. Replacing RemoteCLIP with vanilla CLIP-ViT-B/32 [1] on all 33 conditions, RemoteCLIP wins *every* one (mean Δ mAP $+0.031$, range $+0.014$ to $+0.050$, Table II). The gap is wider for Late ($+0.04$ to $+0.05$) than CA ($+0.014$ to $+0.029$): strong fusion partly compensates for a weaker backbone, but the best run still pairs RemoteCLIP with CA, consistent with Liu et al.’s 6.39% zero-shot advantage.

Per-class decomposition explains where the gain originates: minor classes that the image-only baseline fails on (Shrub F1 = 0.00, Built-up 0.65, Barren 0.61) recover dramatically under CA (Shrub 0.66, Built-up 0.79, Barren 0.79). Captions and images carry complementary information precisely on these classes; the Phase-1 “trivial-task” concern is therefore disproved exactly where it would have mattered.

VI. PRELIMINARY SEGMENTATION EXTENSION

The segmentation pipeline reuses cached patch features with two adaptations: the cross-attention direction is *inverted* (image patches query text tokens), and the head outputs per-patch logits over 7 classes; patch-level ground truth is the majority-pool of the 224×224 mask over each 32×32 patch (a 7×7 grid). We run 7 conditions (image-only + 3 captions $\times$ {late, CA}) with 3 seeds each.

Best CA segmentation reaches mIoU 0.711 ± 0.003 vs. image-only 0.574 ($+0.137$ mIoU, mirroring Path A’s $+0.131$ mAP). CA beats Late on every caption (Fig. 2); the noise-robustness signature persists (Late+vision_qwen mIoU $0.575 \approx$ image-only; CA+vision_qwen 0.610). Per-class IoU shows the largest minor-class lift on Shrub ($0.05 \rightarrow 0.41$,

TABLE II
CROSS-TASK SUMMARY VS. BASELINE. BOTH TASKS SHOW $\sim +0.13$ FUSION GAIN, INDICATING THE CA ADVANTAGE IS NOT TASK-SPECIFIC.

Setting	Baseline	Best CA	Δ
Path A classification (mAP)	0.817	0.948	$+0.131$
Path B segmentation (mIoU)	0.574	0.711	$+0.137$
Backbone (mean mAP, 5 caps)	0.840 (vCLIP)	0.871 (RC)	$+0.031$

$8 \times$), confirming caption guidance contributes spatial localisation signal even where patch-level positives are sparse.

VII. DISCUSSION AND PHASE-3 OUTLOOK

Three findings emerge. First, **CA universally beats alternative fusions** on this benchmark: it wins every head-to-head across $4 \text{ fusion} \times 5 \text{ caption} \times 3 \text{ seed}$ in classification and the seg subset, with statistically negligible variance. Second, **CA’s selectivity yields noise robustness**: vision-only LLM captions hallucinate on this dataset, which Late and Gated cannot filter, but cross-attention’s weights effectively downweight unreliable tokens (CA recovers $+0.049$ mAP on noisy captions where Late gains nothing). Third, **the fusion-gain pattern transfers across tasks**: both classification and segmentation show $\sim +0.13$ absolute gain over image-only baselines (Table II), suggesting the result is not specific to scene-level prediction. The Phase-1 “trivial-task” concern is therefore addressed empirically: per-class, fusion is most valuable precisely on the minor classes captions describe inconsistently (Shrub F1 $0.00 \rightarrow 0.66$; Shrub IoU $0.05 \rightarrow 0.41$).

All gains are with a frozen backbone; the rank order is stable across captions, seeds, and tasks. Phase-3 will extend the seg matrix to all captions $\times 4$ fusions, ablate τ and patch resolution (B/16, 14×14), and explore parameter-efficient fine-tuning.

REFERENCES

- [1] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” *ICML*, 2021.
- [2] C. Wen *et al.*, “Vision-Language Models in Remote Sensing,” *IEEE Geosci. Remote Sens. Mag.*, vol. 12, 2024.
- [3] F. Liu *et al.*, “RemoteCLIP: A Vision Language Foundation Model for Remote Sensing,” *IEEE TGRS*, vol. 62, 2024.
- [4] W. Zhang *et al.*, “EarthGPT: A Universal Multimodal Large Language Model for Multisensor Image Comprehension in Remote Sensing,” *IEEE TGRS*, vol. 62, 2024.
- [5] P. Wang *et al.*, “RingMoGPT: A Unified Remote Sensing Foundation Model for Vision, Language, and Grounded Tasks,” *IEEE TGRS*, vol. 63, 2025.
- [6] Y. Bazi *et al.*, “Bi-Modal Transformer-Based Approach for Visual Question Answering in Remote Sensing Imagery,” *IEEE TGRS*, vol. 60, 2022.
- [7] M. Lan *et al.*, “Language Query-Based Transformer for Visual Grounding on Remote Sensing Images,” *IEEE TGRS*, vol. 62, 2024.
- [8] S. Zhang *et al.*, “SegCLIP: Multimodal Visual-Language and Prompt Learning for RS Semantic Segmentation,” *IEEE TGRS*, vol. 62, 2024.
- [9] E. Perez *et al.*, “FiLM: Visual reasoning with a general conditioning layer,” *AAAI*, 2018.
- [10] L. Chen *et al.*, “ShareGPT4V: Improving Large Multi-Modal Models with Better Captions,” *ECCV*, 2024.
- [11] M. Caglar, “ARAS400K Remote Sensing Dataset,” Zenodo, 2026, <https://zenodo.org/records/18890661>.
- [12] Phase-1 instructor review, DI 725 Term Project, Spring 2026.
- [13] S. Ceran, “ARAS multi-modal attention land-cover classification,” GitHub, 2026, <https://github.com/sceran/aras-multimodal-attention-landcover-classification>.